

A Packing of 10 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 10 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.827986. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.799738, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 10 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.827986 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane’s *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 10$ is blank in Sloane’s table, in Sloane’s coordinate files and in Cohn’s table as checked on October 8, 2026. The best value implied by known packings is 2.799738, obtained by deleting 1 of the 11 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 10/9 = 2.828283$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N10.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author’s laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 62 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
282798648445210190679654277184089170024853954819853476881912246813957011
752307759523264380205925091731169922508002632132173299751179091494905304
4711747363427420022019290311895728758476966454583
```

denominator:

```
999999999999109910312297078730238392392357676084243297648714975750191740
516572724740765101528823885485469574772309470002055710765951087150944100
479555224318211600155687125579569935805481789224
```

B Coordinates

10 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N10.txt`.

```
-586344275528 -341157969877 -152049200071 2823651075237428088057 -128354416940 342721683378 -276396804926968262039 -422655761174 197659631274 296557552565
-137800857929 -183687055629 3767482273086 -81530085628807695562 -272564947542 -348333588124 -1722228519161883067728 -94654587366 118475384139 -421686150873
-13493834800 -358321232541 -47483839651 -1125476799277452424802 -60226299613 -482477986296 -3041640122200050200370 648304316624 220737101010 278625618246
100107149603 -342706381490 -634342686524 -16380594725504847754 -299235733267 191892646400 -41185049993006561237791 -268000734414 -163934205801 -122102482874
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398911190493 259342919578 -214737931836 -1228149377644232153842 61591319871 418517665416 -49258727357 161647465160 368446657337 -238825252565 346075931360
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-177209100914 32932698734 -93157998222 75198026524 -298335186820 184151882414 436683981380 -3506906315158308983643 -227388189915 -260948796804 -360568819632
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